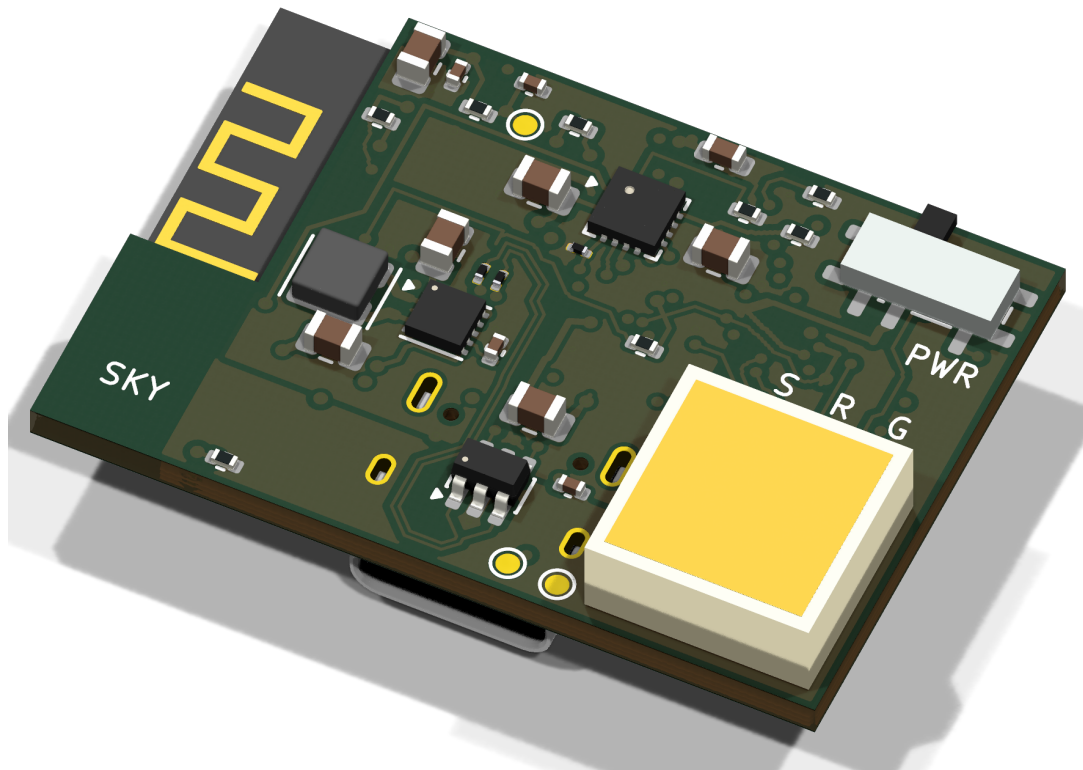


B RID hardware engineering review

Editable KiCad project and actual CAD renders

The 35 x 24 mm nominal hardware layout is routed and passes native ERC, PCB DRC, no-open and schematic-parity checks. Both RF modules now share the final-reflow face. The ceramic patch and all three indicators share the sky-facing face.

Release hold: do not fabricate, assemble or connect a battery using this package as a released specification. Exact battery/NTC/harness qualification, antenna and receiver procurement, process confirmation and physical testing remain open.



Sky-facing side

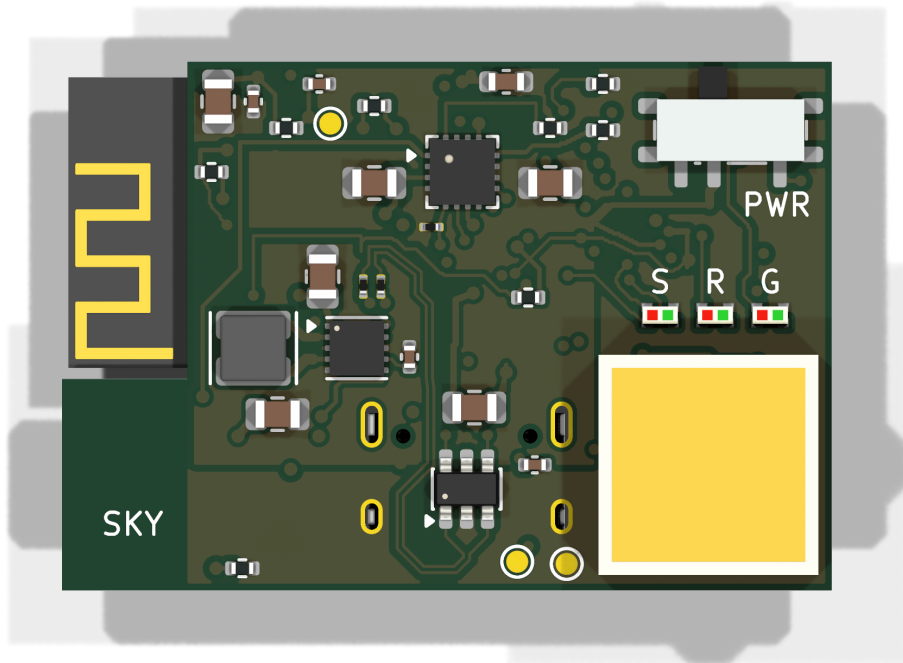
KiCad B side: direct ceramic patch, G / R / S indicators and accessible corner switch. Keep the patch facing open sky in the intended installation. The battery, enclosure and aircraft have not been modeled or qualified.

G = valid GNSS fix; R = actual RID transmit state. Both use green LEDs driven by future firmware.
S-green = enabled 3V3 power; S-red = charger charging indication, including with the system switch off. Simultaneous colors are possible.

These are native-board renders. Several component models are labelled dimensional approximations; LED colors do not show a powered test result.

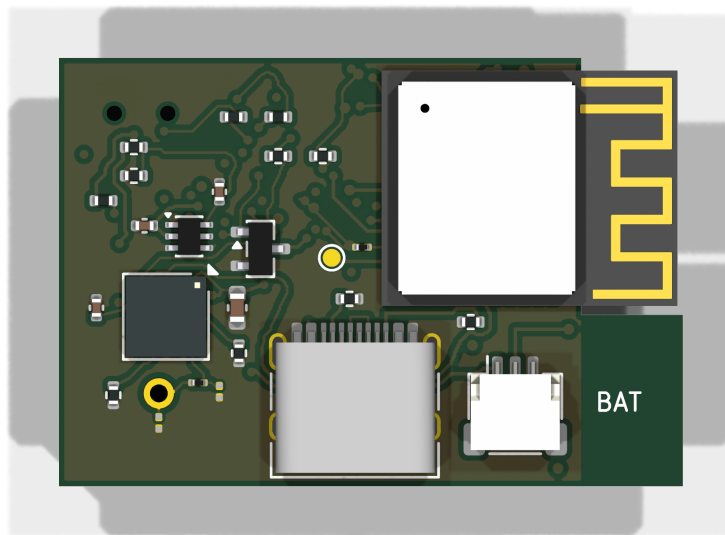
Board faces and mounting

Orthographic CAD views from the current native board



Sky and status face | KiCad B

The SKY marking and G / R / S legends are on this face. The patch body is 10 x 10 x 4 mm nominal and is installed after SMT reflow. SW1 is edge-accessible; verify physical actuator direction before assigning ON/OFF in an enclosure.



Module face | KiCad F

ESP32-C3-MINI-1 and MIA-M10Q are fitted together in the final SMT pass. The ESP antenna extends over the PCB recess while remaining inside the nominal rectangular envelope.

Electrical and geometry verification

Native KiCad 10.0.6 checks and targeted engineering checks

ERC 0 | DRC 0 | opens 0 | parity 0

60 footprint references; 1068 trace segments; 105 filled/capped vias. Four test pads; 56 resolved model references; C13/C14 are DNP. The report includes inherited ignored-check settings and does not call them tested.

USB corridor

The routed pair uses 0.135 mm traces and a 0.150 mm main-corridor gap on the JLC06161H-3313 six-layer basis. Official coplanar and non-coplanar calculations independently support the nominal 90 ohm geometry. F references In1 GND; B references In4 GND. Full-width sampling found no reference loss outside intentional signal-via antipads.

ESD-output-to-MCU copper mismatch is 0.030636 mm. USB-C contact duplication makes complete A-orientation mismatch about 2.079364 mm; the B orientation retains about 0.030636 mm. The longest duplicated-contact branch is 3.11 mm. At full speed, calculated delay is small relative to 4-20 ns transitions; this remains a disclosed compact-layout deviation.

Connector return vias are 0.955 mm from each signal transition. MCU return distances are 1.524 / 2.052 mm, with a second return at 1.636 / 2.145 mm. The ESD ground via is 0.2625 mm from the ground-pad center. These are measured dimensions, not an assertion of ideal placement.

GNSS and power

The new 4.085 mm, 0.150 mm RF feed is on F with In1 GND reference and no signal-layer change. The patch pin requires its intentional plane antipad. GPS L1, Galileo E1 and QZSS L1 are the intended band-compatible modes; the patch specification does not establish all-constellation performance.

Converter bypass/output placements and switching paths were corrected. Exact Murata capacitors were selected using manufacturer bias curves; the two 22 uF output parts screen at about 18.5 uF together at 3.4 V. This engineering margin is not a guaranteed combined process/temperature/aging limit.

The battery ADC now has hardware isolation while 3V3 is off. GPIO strap bias and charging-status isolation are explicit. G/R state control and a valid USB current-grant policy require future firmware; USB SOF traffic alone never authorizes 500 mA.

Manufacturing and remaining gates

What must be resolved before a prototype release

Design checks passed

Native routing/parity checks, pad mapping, 3D dependency resolution, impedance-width basis, targeted reference-plane checks, nominal outline and component-placement checks passed. The verified source and file hashes are included in the project package.

Justified engineering deviations

USB contact-branch asymmetry and measured return distances are documented. The 35 x 24 mm outline is nominal: worst-case module and switch margins are only about 0.05 mm and 0.04 mm using assumed placement tolerances. A hard finished maximum needs agreed profile and assembly tolerances.

Supplier and part gates

Battery: choose an exact protected 1S pack with verified continuous/pulse current, maximum charge-voltage tolerance, NTC curve, charge-temperature window, safety-timer compatibility and keyed harness. The former 500 mAh target is not a qualified pack. Compact public leads lack enough data; the larger 1 Ah fallback would materially enlarge the device.

GNSS: confirm genuine supply of MIA-M10Q-00B and INPAQ PA1575MQ4S-123-1Z. Exact antenna stock was not verified. Confirm the engineering-derived pin footprint, adhesive and ground interface. Its 20 x 20 mm reference ground is different from this host.

Process: B-side ordinary parts first, F-side modules last, ceramic patch after SMT. Confirm all selected parts tolerate that profile and obtain via-fill/cap, ENIG, finished-stackup and impedance acceptance. Ordinary tenting is not a substitute for filled/capped via-in-pad.

Bench and functional gates

Measure GNSS matching/acquisition/coexistence; USB enumeration, both orientations, signal integrity and ESD; charging/current/temperature/timer behavior; power startup, droop and thermal limits; ADC off-state isolation; LED visibility and mechanical fit. MCU-off host-suspend detection is unresolved. No firmware, broadcast, RF compliance or flight qualification is claimed.

Next action

No additional placement decision is needed to use the reviewed editable CAD. A battery specification and supplier/process confirmation are needed before fabrication; any supplier outreach or order requires separate authorization. Manufacturing CAM and assembly BOM/CPL are intentionally withheld.

Source identity and references

Traceable source files and primary technical references

Verified source

PCB SHA256

b850f5ccaf6a7c0b230d44fd47341316aeb6269b236a244414ebca5368a36045

The original transferred baseline is retained separately. Sync-back must compare the Mac files with that baseline and preserve any user changes and unsaved editor state. The package contains the current editable project, local symbols/footprints/models, native reports, detailed engineering notes and actual renders.

Key primary sources

[Espressif module and layout](#)

[Espressif USB guidance](#)

[u-blox MIA integration](#)

[INPAQ exact antenna engineering specification](#)

[JLC impedance calculator](#)

[TI charger datasheet](#)

[TI converter datasheet](#)

[Murata capacitor characteristics](#)

[Kingbright LED datasheet](#)

[SHOU HAN switch drawing](#)

Detailed sources, point-in-time stock observations, exact component selections and limitations are in the included engineering notes. Approximate 3D models are visual aids and do not constitute vendor mechanical approval.